GAURAV P. LUTE

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Location :- Pune, Maharashtra

SUMMARY

Professional with 1 years of experience in machine learning Model Development in the Loancare Domain. Experienced in working with US Mortgage datasets to generate market insights for loancare. Proficient in Python, SQL and R, with a strong ability to deliver actionable insights that support strategic decision-making in financial institutions. Experienced in working with real-world datasets and improving ML model performance. Actively seeking full-time opportunities in Data Analyst, Data Science, Machine Learning, and AI-driven analytics.

SKILLS & PROFICIENCIES

Technical Skills:-

- **Python** :- Numpy, Pandas, Scikit-learn, Data preprocessing, Tensorflow, PyTorch, Pydantic, OpenAi.
- **SQL/R** :- Data Cleaning, Transformation, Extraction and data visualization.
- **Statistics** :- Hypothesis Testing, Z test, T test and EDA.
- **Time series models** :- ARIMA and Prophet.
- **Machine Learning Algorithms**:- Linear and Logistics Regression , Random Forest and Xgboost.
- **Deep Learning Algorithms** :- Convolution Neural Network (CNN) and LSTM (Long and Short term memory)
- **LLM and GenAI** :- OpenAi, pydantic and prompt engineering.

EXPERIENCE

Fidelity National Finance - Trainee **Sep 2025 – Mar 2026**
LLM Extraction and Pattern Analysis. **Bangalore**

- Developed a **ChatGPT-based extraction system using the OpenAI API**, implementing **JSON schema validation with Pydantic classes** to accurately extract structured fields from **financial and mortgage-related documents or OCR Text.**
- Designed and implemented **prompt optimization strategies** to improve extraction accuracy and consistency in LLM outputs.
- Performed **pattern analysis on LLM failure cases** to identify model limitations and contributed to the development of improved prompts and validation strategies.
- Ensured reliable structured outputs by combining schema validation, prompt engineering, and iterative evaluation techniques.

Spatialty.Ai - Data Science Intern **Feb 2025 – Jun 2025**
Machine Learning Models. **Bangalore**

- Developed a solar irradiance prediction **machine learning model** using 10 years of satellite data.
- Conducted EDA and Trained **Linear Regression and Random Forest models**, achieving **$R^2 = 0.84$**.
- Built time series forecasting models to predict March 2025 solar irradiance with MAPE = 3%.
- Optimized forecasts using **XGBoost** and **Prophet**, achieving the lowest MAPE of 0.03.

PERSONAL PROJECTS

- **Credit Risk Modeling | |** Link] :- Developed a machine learning model to predict loan repayment probability. Performed EDA, data cleaning, and feature engineering on financial data. Trained **Logistic Regression,** achieving approximately **70% accuracy.**
- **Fake Image Detection | |** Link] :- Developed a **CNN-based deep learning model t**o classify real vs AI-generated images.Preprocessed image datasets and optimized training pipelines. Evaluate a model accuracy achieve 80%.
- **Next Word Prediction | |** Link] :- Developed an LSTM-based language model for next-word prediction. Applied **tokenization** and **sequence padding** on large text corpora. Implemented real-time user input for interactive predictions.

EDUCATION

- **M.Tech in modeling and simulation | SPPU Pune - June 2025** CGPA :- 7.56
  Savitribai Phule Pune University, Pune.
- **B.E in Automobile Engineering | SPPU Pune - June 2023** CGPA :- 8.33
  Governmet College of Engg and Research Avsari Kh, Pune.
- **DIPLOMA in Automobile Engineering - June 2020** Percentage :-93.10%
  Governmet Polytechnic Nagpur.

CERTIFICATE

- **Credit Risk Modeling in python - Udemy | Link**
- **Python Certificate - HackerRank. | Link**